When constant g is not enough

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Recognize the limits of the near-surface mgh model.

SUCCESS CRITERIA - I can...

1. Compare altitude with planetary radius.
2. Explain that gravitational field weakens with center distance.
3. Choose a model appropriate to the height interval.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “assumption” and “field” to explain your model or result. Sentence starters are optional.

assumption: a condition accepted for using a model

field: a description of an influence at each location

potential energy: energy associated with the configuration of interacting objects

01 Bring back an earlier idea

A 3 kg cart starts at rest. A 19.5 N pull and 7 N resistance act over 3 m.
Find final speed.

Word help: final = at the end of the interval; subscript f

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta U \approx mg\Delta h$; $g = GM/r^2$.
The first relation needs nearly constant g. For large changes in r, the field changes; one fixed surface value is insufficient.

Worked reasoning: A local model treats g as constant over small height changes. At a center distance twice a planet's radius, however, g is only one quarter of its surface value.

Your turn: Sketch a planet and mark center distances R and 2 R.
Explain why one surface g cannot describe the entire trip exactly.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Compare lifting a book 1 m with moving a satellite one planetary radius above the surface. Which better fits constant g and why?

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

04 Solve and justify

For a planet with $GM=100\ m^3/s^2$, mass 2 kg moves from r=10 to r=20 m. Find $\Delta U=GMm\ (1/r_i-1/r_f)$.
Compare surface-g estimate mgh using $g=1\ m/s^2$ and h=10 m.

Word help: estimate = find an approximate value

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why is altitude h not interchangeable with center-to-center r in the universal expression?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Total momentum initially is (24, 0) kg·m/s. One fragment leaves with (4, 6) kg·m/s.
Find the other momentum vector and magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Using the same toy planet, compare gravity at r=10 and 20 m. Connect the result to the energy-model choice.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A planet has GM=100 m^3/s^2 and radius 5 m.
Compare exact gravitational potential-energy increase for a 2 kg object raised to r=10 m with the surface-g approximation.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3